

Vireen Chowdary Vesangi

Applied ML | Data Engineering | Full-Stack Systems

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SUMMARY

Engineer combining **applied machine learning**, **probabilistic modelling**, and **production system design**. B.Tech in **Artificial Intelligence Engineering** (Robotics specialization) plus MBA in Information Systems & Analytics. Hands-on with the full ML lifecycle: **ETL pipelines** over hundreds of thousands of records, **empirical-Bayes** and **hierarchical shrinkage** models, **Monte Carlo simulation**, model **calibration & validation**, and **FastAPI + React** deployment on cloud (Render/Vercel). Comfortable owning problems end-to-end — from raw data through statistical modelling to a live, user-facing product.

TECHNICAL SKILLS

Languages: **Python** (Pandas, NumPy, scikit-learn, Matplotlib, Seaborn), **SQL** (joins, aggregations, subqueries, query optimization), R (basics), C/C++ (academic).

Machine Learning & Statistics: **Probabilistic & generative modelling**, **empirical-Bayes shrinkage**, **hierarchical / multi-level modelling**, **Monte Carlo simulation**, **feature engineering**, model **calibration & validation**, hypothesis testing, regression (Linear, Logistic), classification (**KNN**, Random Forest), **clustering (K-Means)**, time-series forecasting (Prophet), **NLP (BERT)**, statistical testing (ANOVA, MANOVA, Chi-Square, t-test, Mann-Whitney U).

Data Engineering: **ETL pipelines**, JSON parsing, groupby aggregation, categorical bucketing, feature engineering, data cleaning & validation, multi-source data reconciliation.

Backend & Frameworks: **FastAPI**, **stateless architecture**, **pytest** (unit, invariant, regression tests), Streamlit, Pydantic schemas, Ollama (local LLM).

Frontend & DevOps: Component/state design, reactive UI, Git / GitHub, cloud deployment (**Vercel**, **Render**), environment configuration, CORS, analytics instrumentation.

BI & Visualization: **Power BI** (DAX, Power Query, Data Modelling, RLS), **Tableau** (LOD expressions, parameters), Advanced Excel (Pivot Tables, VLOOKUP, VBA), SPSS.

EXPERIENCE

Data Analyst Intern (Reporting & Analytics)

May – Jul 2025

Symbiosys Technologies, Visakhapatnam

- Gathered and prepared data from multiple internal sources (client database, invoice exports, web forms) covering **1,500+ multi-currency records** across verticals; performed end-to-end analysis and presented results to leadership.
- Wrote **SQL queries** (joins, aggregations, subqueries) and used **Python** and Advanced Excel to **clean, transform, and validate** data; developed an automated Python currency-conversion workflow (with fallback API handling) to standardize multi-currency invoices for consistent reporting.
- Built **3 Tableau dashboards** tracking revenue, overdue payments, and quarterly performance across **~Rs 2.7 Cr in total overdue exposure**; isolated **~Rs 1.2 Cr** concentrated in a single high-risk market and a single client representing **~Rs 72 L** in receivables. **Spearman rank correlation** ($\rho = -0.001$, $p = 0.989$) confirmed invoice size was **not** a predictor of payment delay — shifting management attention from invoice value to client-behaviour patterns.

Artificial Intelligence Intern

Feb – May 2023

Ozonetel Communications Pvt. Ltd., Hyderabad

- Built a front-end web interface for an Agent Assist AI product, handling integration between backend (FAST API) and the customer-facing UI.
- Applied **BERT-based NLP** modelling to unstructured customer-interaction data and contributed to model-validation workflows in a production business context.

PORTFOLIO PROJECTS

Captain's Call — Probabilistic IPL T20 Match Simulator (Solo, Deployed) *Python, FastAPI, React, pandas*

- Designed, built, and **deployed end-to-end** a live, data-driven IPL cricket match simulator (Python/FastAPI backend + React/Vite frontend on Render/Vercel), owning **data engineering, statistical modelling, backend API, frontend UI, and deployment solo**.
- Engineered a **pandas ETL pipeline** transforming raw nested Cricsheet JSON for **1,244 matches / 295,733 ball-by-ball deliveries** across 18 IPL seasons (2008–2026) into analysis-ready datasets via parsing, groupby aggregation, feature engineering, and categorical bucketing across **4 eras and 7 innings sub-phases**.
- Developed a **probabilistic outcome model** estimating per-ball distributions over **6 canonical outcomes** conditioned on batter, bowler, era, and phase, using a **multiplicative baseline $\times$ player-multiplier** formulation with **empirical-Bayes shrinkage** ($k=20$).
- Implemented **multi-level hierarchical shrinkage** (sub-phase $\rightarrow$ phase $\rightarrow$ player $\rightarrow$ role-group $\rightarrow$ league) across a

- 26,250-cell parameter grid per discipline**; a sparsity audit over 2,765 audited cells confirmed **~75% data-rich** and only ~10–12% below shrinkage threshold.
- Built a **Monte Carlo win-probability engine (2,000 simulations per match)**; **diagnosed and fixed a systematic 64/36 estimation bias** caused by short-order replay, restoring ~50/50 symmetry for identical line-ups — verified empirically.
 - Designed a **role-aware fallback** with a **data-scaled miscast penalty** that closed a gameplay exploit and made out-of-position players behave realistically; **calibrated** the intent-control layer to **MAE ~1.04pp** against real attacking-situation data.
 - Architected a **stateless, one-ball-per-call FastAPI backend (state-in / state-out)**, structured-event outputs) enabling a pure-function React UI, server-side rule enforcement, and horizontal scalability; wrote a **GPU-free pytest suite** (invariant, monotonicity, and symmetry regression tests).
 - Shipped to production with **usage analytics** and an in-app feedback pipeline; published a public showcase repo (README + live demo) with model and data held privately.
 - [Click here to play with the Simulation](#)
 - [Click here to view the GitHub repo](#)

Pilot-to-Scale Decision Support for Autonomous Robotics

Python, Ollama, Streamlit, Power BI

- Built a **local-LLM analytics copilot** (Python, Streamlit, **Ollama**, **Pydantic**-validated schemas) that converts unstructured operational text — pilot notes, incident reports — into structured, validated analytical artifacts, demonstrating applied **GenAI integration into BI workflows**.
- Complemented with a multi-table **Power BI dashboard** (5 fact / dimension tables, 4 pilots) featuring **DAX measures**, weighted readiness scoring, and fleet-level rollups for decision support.

ACADEMIC PROJECTS

GeoSpot — Multi-modal image geolocation system *Python, PyTorch, OpenCLIP, YOLOv8, EasyOCR, Gradio, uv*

- Designed and shipped a **multi-modal** country-prediction pipeline fusing three deep-learning modules (**OpenCLIP scene**, **EasyOCR text**, **YOLOv8 flag detection**) with trust-weighted fusion, achieving 73.9% top-1 and 87.0% top-3 accuracy across 57 countries on a hand-labeled 23-image test set.
- Fine-tuned **YOLOv8n** on 4,200+ synthetic "flag-in-scene" composites (28 countries, cut-and-paste augmentation) after diagnosing that naive whole-image training produced degenerate predictions; achieved 99.3% validation mAP@0.5.
- Replaced hand-picked fusion weights with a trust-weighted scheme calibrated from measured per-module precision, improving end-to-end top-1 accuracy from 66.7% to 73.9%.
- Built a Gradio web demo with GPU-accelerated inference (2s/image on RTX 4060 Ti), reproducible via **uv** sync with **CUDA-pinned PyTorch**; 5-test pytest suite runs in CI without model downloads.
- [Click here to view the GitHub repo](#)

AI-Driven Forecasting & Customer Segmentation

Prophet, Random Forest, K-Means

- Built time-series models (Prophet, Random Forest) for revenue and occupancy forecasting; deployed K-Means segmentation to classify customer cohorts and define revenue KPIs.

AR/VR Consumer Study — Statistical Research (BRM)

SPSS, Python, NLP

- Designed and executed a mixed-methods consumer research study (**134 survey responses** + retailer interviews) evaluating how AR/VR experiences shape brand loyalty, trust, and customer response.
- Tested **4 hypotheses** using **MANOVA**, **multiple regression** (Brand Loyalty model $R^2 = 0.448$, Trust strongest predictor), and **Mann-Whitney U**; complemented with **Python NLP keyword-frequency analysis** on retailer transcripts to surface qualitative adoption barriers.

EDUCATION

MBA — Information Systems, Analytics & International Business <i>GITAM University, Visakhapatnam</i>	2024 – 2026 <i>CGPA: 7.6/10</i>
B.Tech — Artificial Intelligence Engineering (Robotics Specialization) <i>Amrita Vishwa Vidyapeetham, Amritapuri</i>	2019 – 2023 <i>CGPA: 7.5/10</i>

CERTIFICATIONS

- Microsoft PL-300 Power BI Data Analyst (in progress)
- Microsoft Power BI Data Analyst (Coursera)
- Microsoft Engage 2021
- Shore Fest Datathon 1st Place Finish